Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

November 16, 2001, was a Friday. If each of the years 2004, 2008, and 2012 had 366 days, and the remaining years from 2001 through 2014 had 365 days, what day of the week was November 16, 2014?

A. Sunday

B. Monday

C. Tuesday

D. Wednesday

E. Thursday

•

How many prime numbers between 1 and 100 are factors of 7,150?

A. One

B. Two

C. Three

D. Four

E. Five

•

A sequence of numbers $a_1, a_2, a_3, \ldots$ is defined as follows: $a_1 = 3$, $a_2 = 5$, and every term in the sequence after a_2 is the product of all terms in the sequence preceding it, e.g., $a_3 = (a_1)(a_2)$ and $a_4 = (a_1)(a_2)(a_3)$. If $a_n = t$ and $n > 2$, what is the value of a_{n+2} in terms of t?

A. $4t$

B. t^2

C. t^3

D. t^4

E. t^8

•

Last year the price per share of Stock X increased by k percent and the earnings per share of Stock X increased by m percent, where k is greater than m. By what percent did the ratio of price per share to earnings per share increase, in terms of k and m?

A. $\frac{k}{m}\%$

B. $(k-m)\%$

C. $\frac{100(k-m)}{100+k}\%$

D. $\frac{100(k-m)}{100+m}\%$

E. $\frac{100(k-m)}{100+k+m}\%$

•

Of the 300 subjects who participated in an experiment using virtual-reality therapy to reduce their fear of heights, 40 percent experienced sweaty palms, 30 percent experienced vomiting, and 75 percent experienced dizziness. If all of the subjects experienced at least one of these effects and 35 percent of the subjects experienced exactly two of these effects, how many of the subjects experienced only one of these effects?

A. 105

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

A. 105

B. 125

C. 130

D. 180

E. 195

•

If $m^{-1} = -\frac{1}{3}$, then m^{-2} is equal to

A. -9

B. -3

C. $-\frac{1}{9}$

D. $\frac{1}{9}$

E. 9

•

A photography dealer ordered 60 Model X cameras to be sold for $250 each, which represents a 20 percent markup over the dealer's initial cost for each camera. Of the cameras ordered, 6 were never sold and were returned to the manufacturer for a refund of 50 percent of the dealer's initial cost. What was the dealer's approximate profit or loss as a percent of the dealer's initial cost for the 60 cameras?

A. 7% loss

B. 13% loss

C. 7% profit

D. 13% profit

E. 15% profit

•

Seven pieces of rope have an average (arithmetic mean) length of 68 centimeters and a median length of 84 centimeters. If the length of the longest piece of rope is 14 centimeters more than 4 times the length of the shortest piece of rope, what is the maximum possible length, in centimeters, of the longest piece of rope?

A. 82

B. 118

C. 120

D. 134

E. 152

•

What is the difference between the sixth and the fifth terms of the sequence 2, 4, 7, … whose n^{th} term is $n + 2^{n-1}$?

A. 2

B. 3

C. 6

D. 16

E. 17

•

From the consecutive integers –10 to 10, inclusive, 20 integers are randomly chosen with repetitions allowed. What is

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

From the consecutive integers –10 to 10, inclusive, 20 integers are randomly chosen with repetitions allowed. What is the least possible value of the product of the 20 integers?

A. $(-10)^{20}$

B. $(-10)^{10}$

C. 0

D. $-(10)^{19}$

E. $-(10)^{20}$

•

The letters D, G, I, I, and T can be used to form 5-letter strings such as DIGIT or DGIIT. Using these letters, how many 5-letter strings can be formed in which the two occurrences of the letter I are separated by at least one other letter?

A. 12

B. 18

C. 24

D. 36

E. 48

•

$$\frac{0.99999999}{1.0001} - \frac{0.99999991}{1.0003} =$$

A. 10^{-8}

B. $3(10^{-8})$

C. $3(10^{-4})$

D. $2(10^{-4})$

E. 10^{-4}

•

Last Sunday a certain store sold copies of Newspaper A for \$1.00 each and copies of Newspaper B for \$1.25 each, and the store sold no other newspapers that day. If r percent of the store's revenue from newspaper sales was from Newspaper A and if p percent of the newspapers that the store sold were copies of Newspaper A, which of the following expresses r in terms of p?

A. $\frac{100p}{125-p}$

B. $\frac{150p}{250-p}$

C. $\frac{300p}{375-p}$

D. $\frac{400p}{500-p}$

E. $\frac{500p}{625-p}$

•

For the past n days, the average (arithmetic mean) daily production at a company was 50 units. If today's production of 90 units raises the average to 55 units per day, what is the value of n?

DO NOT COPY
filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

of 90 units raises the average to 55 units per day, what is the value of n?

A. 30

B. 18

C. 10

D. 9

E. 7

•

If a two-digit positive integer has its digits reversed, the resulting integer differs from the original by 27. By how much do the two digits differ?

A. 3

B. 4

C. 5

D. 6

E. 7

•

In an electric circuit, two resistors with resistances x and y are connected in parallel. In this case, if r is the combined resistance of these two resistors, then the reciprocal of r is equal to the sum of the reciprocals of x and y. What is r in terms of x and y?

A. xy

B. $x+y$

C. $\frac{1}{x+y}$

D. $\frac{xy}{x+y}$

E. $\frac{x+y}{xy}$

•

Xavier, Yvonne, and Zelda each try independently to solve a problem. If their individual probabilities for success are $\frac{1}{4}$, $\frac{1}{2}$, and $\frac{5}{8}$, respectively, what is the probability that Xavier and Yvonne, but not Zelda, will solve the problem?

A. $\frac{11}{8}$

B. $\frac{7}{8}$

C. $\frac{9}{64}$

D. $\frac{5}{64}$

E. $\frac{3}{64}$

•

If $\frac{1}{x} - \frac{1}{x+1} = \frac{1}{x+4}$, then x could be

A. 0

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

A. 0

B. –1

C. –2

D. –3

E. –4

•

$$\left(\frac{1}{2}\right)^{-3}\left(\frac{1}{4}\right)^{-2}\left(\frac{1}{16}\right)^{-1} =$$

A. $\left(\frac{1}{2}\right)^{-48}$

B. $\left(\frac{1}{2}\right)^{-11}$

C. $\left(\frac{1}{2}\right)^{-6}$

D. $\left(\frac{1}{8}\right)^{-11}$

E. $\left(\frac{1}{2}\right)^{-6}$

•

List T consists of 30 positive decimals, none of which is an integer, and the sum of the 30 decimals is S. The estimated sum of the 30 decimals, E, is defined as follows. Each decimal in T whose tenths digit is even is rounded up to the nearest integer, and each decimal in T whose tenths digit is odd is rounded down to the nearest integer; E is the sum of the resulting integers. If $\frac{1}{3}$ of the decimals in T have a tenths digit that is even, which of the following is a possible value of $E - S$?

I. –16

II. 6

III. 10

A. I only

B. I and II only

C. I and III only

D. II and III only

E. I, II, and III

•

If $5 - \frac{6}{x} = x$, then x has how many possible values?

A. None

B. One

C. Two

D. A finite number greater than two

E. An infinite number

DO NOT COPY
filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

•

Seed mixture X is 40 percent ryegrass and 60 percent bluegrass by weight; seed mixture Y is 25 percent ryegrass and 75 percent fescue. If a mixture of X and Y contains 30 percent ryegrass, what percent of the weight of the mixture is X?

A. 10%

B. $33\frac{1}{3}\%$

C. 40%

D. 50%

E. $66\frac{2}{3}\%$

•

How many of the integers that satisfy the inequality $\frac{(x+2)(x+3)}{x-2} \geq 0$ are less than 5?

A. 1

B. 2

C. 3

D. 4

E. 5

•

Of the 150 houses in a certain development, 60 percent have air-conditioning, 50 percent have a sunporch, and 30 percent have a swimming pool. If 5 of the houses have all three of these amenities and 5 have none of them, how many of the houses have exactly two of these amenities?

A. 10

B. 45

C. 50

D. 55

E. 65

•

The value of $\frac{2^{-14}+2^{-15}+2^{-16}+2^{-17}}{5}$ is how many times the value of 2^{-17}?

A. $\frac{3}{2}$

B. $\frac{5}{2}$

C. 3

D. 4

E. 5

•

Which of the following fractions has a decimal equivalent that is a terminating decimal?

A. $\frac{10}{189}$

B. $\frac{15}{}$

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

B. $\frac{15}{196}$

C. $\frac{16}{225}$

D. $\frac{25}{144}$

E. $\frac{39}{128}$

•

If $\left(\frac{1}{5}\right)^m \left(\frac{1}{4}\right)^{18} = \frac{1}{2(10)^{35}}$, then $m =$

A. 17

B. 18

C. 34

D. 35

E. 36

•

From a group of 8 volunteers, including Andrew and Karen, 4 people are to be selected at random to organize a charity event. What is the probability that Andrew will be among the 4 volunteers selected and Karen will not?

A. $\frac{3}{7}$

B. $\frac{5}{12}$

C. $\frac{27}{70}$

D. $\frac{2}{7}$

E. $\frac{9}{35}$

•

An investor purchased 100 shares of Stock X at $6\frac{1}{18}$ dollars per share and sold them all a year later at 24 dollars per share. If the investor paid a 2 percent brokerage fee on both the total purchase price and the total selling price, which of the following is closest to the investor's percent gain on this investment?

A. 92%

B. 240%

C. 280%

D. 300%

E. 380%

•

If x is the product of the integers from 1 to 150, inclusive, and 5^y is a factor of x, what is the greatest possible value of y?

A. 30

B. 34

DO NOT COPY
filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

B. 34

C. 36

D. 37

E. 39

•

The number of defects in the first five cars to come through a new production line are 9, 7, 10, 4, and 6, respectively. If the sixth car through the production line has either 3, 7, or 12 defects, for which of these values does the mean number of defects per car for the first six cars equal the median?

I. 3

II. 7

III. 12

A. I only

B. II only

C. III only

D. I and III only

E. I, II, and III

•

A furniture dealer purchased a desk for $150 and then set the selling price equal to the purchase price plus a markup that was 40 percent of the selling price. If the dealer sold the desk at the selling price, what was the amount of the dealer's gross profit from the purchase and the sale of the desk?

A. $40

B. $60

C. $80

D. $90

E. $100

•

How many odd numbers between 10 and 1,000 are the squares of integers?

A. 12

B. 13

C. 14

D. 15

E. 16

•

Three grades of milk are 1 percent, 2 percent, and 3 percent fat by volume. If x gallons of the 1 percent grade, y gallons of the 2 percent grade, and z gallons of the 3 percent grade are mixed to give $x + y + z$ gallons of a 1.5 percent grade, what is x in terms of y and z?

A. $y + 3z$

B. $\frac{y+z}{4}$

C. $2y + 3z$

D. $3y + z$

E. $3y + 4.5z$

•

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

•

There are 8 books on a shelf, of which 2 are paperbacks and 6 are hardbacks. How many possible selections of 4 books from this shelf include at least one paperback?

A. 40

B. 45

C. 50

D. 55

E. 60

•

If $M = \sqrt{4} + \sqrt[3]{4} + \sqrt[4]{4}$, then the value of M is

A. less than 3

B. equal to 3

C. between 3 and 4

D. equal to 4

E. greater than 4

•

When a certain tree was first planted, it was 4 feet tall, and the height of the tree increased by a constant amount each year for the next 6 years. At the end of the 6th year, the tree was $\frac{1}{5}$ taller than it was at the end of the 4th year. By how many feet did the height of the tree increase each year?

A. $\frac{3}{10}$

B. $\frac{2}{5}$

C. $\frac{1}{2}$

D. $\frac{2}{3}$

E. $\frac{6}{5}$

•

A certain musical scale has 13 notes, each having a different frequency, measured in cycles per second. In the scale, the notes are ordered by increasing frequency, and the highest frequency is twice the lowest. For each of the 12 lower frequencies, the ratio of a frequency to the next higher frequency is a fixed constant. If the lowest frequency is 440 cycles per second, then the frequency of the seventh note in the scale is how many cycles per second?

A. 440 $\sqrt{2}$

B. 440 $\sqrt{2^7}$

C. 440 $\sqrt{2^{12}}$

D. 440 $\sqrt[12]{2^7}$

E. 440 $\sqrt[7]{2^{12}}$

DO NOT COPY
filippo.fiz@uptoten.it